

1. Fill up the blanks in the following statements:

1×5=5

- (a) Image correlation and convolution operations are called _____ operation.
- (b) The _____ filter is a linear smoothing filter.
- (c) Point processing operates on a _____ of the input image.
- (d) Sum of the weights of a smoothing filter is _____.
- (e) Gamma correction is done using _____ transformation.

2. Answer the following questions:

2×4=8

- (a) Show a one-dimensional binomial smoothing filter with a width of seven pixels.
- (b) What single filter is equivalent to applying the following filter twice? (Hint: Convolute the filter with itself).

1	2	1
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- (c) Present mathematical expressions for (i) correlation and (ii) convolution of image $f(x, y)$ with filter $w(s, t)$.
- (d) Find the eight neighbours of pixel p located at coordinates (x, y) . Consider that the origin of the image is located at the top left corner of the image.

3. Consider the following 3×6 image with eight possible gray levels in the range [0, 7].

3

6	2	3	1	2	1
0	6	5	1	6	0
5	1	7	6	1	2

Construct the gray level histogram and the thresholded image obtained using the pixel value of the minimum point between the two largest histogram peaks as the threshold value.

4. A dark low contrast image is given to you. How do you convert it to a good image with high contrast? Justify your answer.

3

5. Differentiate between Box filter and Binomial filter for image smoothing.

2

6. Perform convolution of the 2×4 image given below with the 1×3 row filter given to its right.

3

5	1	4	6
4	7	2	3

1	2	3
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7. Apply a 3×3 median filter to the following image. Consider the output values at the boundary pixels to be undefined.

3

5	1	4	6
4	7	2	3
1	2	3	5

P.T.O.